

## ANSWERS

1.

| Question | Answer                                                                                                                                                                                                                      | Marks |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1(a)     | (a vector quantity) / it has a direction                                                                                                                                                                                    | B1    |
| 1(b)     | <b>acceleration</b> and <b>momentum</b> underlined <b>and</b> no others                                                                                                                                                     | B1    |
| 1(c)     | <b>speed:</b> ( $v = \sqrt{6.8^2 + 2.4^2}$ <b>or</b> $v^2 = 6.8^2 + 2.4^2$ <b>or</b> correct triangle / rectangle drawn / intersecting arcs (by eye)                                                                        | B1    |
|          | <b>speed:</b> 7.1 – 7.3 (m / s)                                                                                                                                                                                             | B1    |
|          | <b>angle:</b> ( $\theta = \tan^{-1}(6.8 / 2.4)$ <b>or</b> $\tan \theta = 6.8 / 2.4$ <b>or</b> 18 – 21(°) <b>or</b> correct triangle / rectangle drawn / intersecting arcs <b>and</b> scale <u>used</u> $\geq 1$ cm: 1 m / s | B1    |
|          | <b>angle:</b> 69 – 72(°)                                                                                                                                                                                                    | B1    |

2.

| Question  | Answer                                                                                                         | Marks |
|-----------|----------------------------------------------------------------------------------------------------------------|-------|
| 7(a)(i)   | 85 m                                                                                                           | B1    |
| 7(a)(ii)  | (speed =) $d / t$ in any form algebraic or numerical                                                           | C1    |
|           | 2.1 m / s                                                                                                      | A1    |
| 7(b)(i)   | displacement involves direction <b>or</b> is a vector                                                          | B1    |
| 7(b)(ii)  | 5 m shown on dotted line                                                                                       | B1    |
| 7(b)(iii) | length of diagonal 9.2 (cm) <span style="float: right;"><math>\sqrt{30 \times 30 + 35 \times 35}</math></span> | B1    |
|           | 46 m <span style="float: right;">46 m</span>                                                                   | B1    |
| 7(b)(iv)  | 39–42° <b>or</b> 318–320°                                                                                      | B1    |

3.

| Question | Answer                                                                                   | Marks |
|----------|------------------------------------------------------------------------------------------|-------|
| 1(a)     | scale vector diagram showing 12 N, 36 N and correct resultant force in correct direction | B1    |
|          | 38 $\pm$ 1 kN                                                                            | B1    |
|          | 018 $\pm$ 3°                                                                             | B1    |

4.

| Question | Answer                                                                                                                               | Marks |
|----------|--------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1(a)(i)  | vector quantity has direction / scalar quantity does not have direction                                                              | B1    |
| 1(a)(ii) | any two scalar quantities, e.g. mass, distance / length, time, speed                                                                 | B1    |
|          | any two vector quantities, e.g. velocity, acceleration, displacement, force / weight                                                 | B1    |
| 1(b)     | P and Q drawn in correct directions to meet at a point with correct size                                                             | B1    |
|          | triangle or parallelogram shown with all three vectors in correct directions <b>and</b> the resultant clear (e.g. as another vector) | B1    |

- 5.
- (a) (i) (vector) has direction **or** scalar does not have direction **or**  
(vectors) may cancel **or** scalars cannot cancel B1
- (ii) one vector quantity e.g.: displacement; velocity, acceleration B1  
one scalar quantity e.g.: distance, length, speed, time, current, resistance B1 [3]
- (b) downward weight arrow of length 3.9–4.1 cm B1  
correct triangle/parallelogram drawn **and** correct diagonal clear B1  
270–285 kN B1  
**horizontal** ( $\pm 3^\circ$ ) B1
- 6.
- (a) (i) arrow(head) on chain pointing to the right B1
- (ii) vertical arrow downwards and part of arrow touching **or** within rectangle  
of lights **or** direction of arrow in (i) **and** (ii) correct (by eye) B1
- (b) scale given (must have unit of cm:N **or** cm/N **or** N:cm **or** N/cm) B1  
correct triangle **or** rectangle (might be implied) and correct resultant  
(compulsory e.c.f. from (i) **or** (ii): i.e. correct diagonal according to  
candidate's (i) and (ii)) B1  
 $272 \leq \text{candidate's value} \leq 283 \text{ N}$  B1 [5]
- 7.
- (a) 950 N B1  
upwards B1
- (b) correct rectangle **and** diagonal **and** at least one velocity labelled  
**or** correct triangle and at least one velocity labelled  
(either way round) B1  
from 7.8(0000) to 8.0(0000) m/s (inclusive) B1  
scale stated B1 [5]
- 8.
- (a) components shown on correct diagram with correct resultant (i.e. towards NE) and  
a scale given B1  
540 ( $\pm 10$ ) m B1  
 $22^\circ \pm 3^\circ$  E of N with correct diagonal B1 [3]
- (b) idea that ends at start, returns in opposite direction B1 [1]
- 9.
- (a) diagram of two forces **and** resultant B1  
W / 6(N) **and** T / 8(N) marked on perp. forces **or** scale given B1  
10(.0  $\pm 0.2$ ) N B1  
35–39° from T/Y/horizontal or 51–55° from W/vertical **and** correct resultant B1
- (b) 10(.0) N **or** e.c.f. B1 [5]